

Home Lab Rack Build Plan

15U black/white rack concept for media, firewall, Ubuntu/Windows compute, and ML trading workloads

Decision Summary

Use the QNAP TS-433eU-US as a compact 1U storage appliance, not as the main compute server. Use the two reused PCs in Sliger CX4170a cases for GPU compute, gaming, Docker services, and trading workloads. Use a dedicated OPNsense firewall box rather than putting routing on the NAS or a gaming PC.

- Keep the NAS, firewall, and compute roles separate so failures and updates do not take down everything.
- Upgrade RAM and NVMe before buying more GPUs; the current CPUs/GPUs are already useful.
- If using CX4170a cases, buy a rack with real rear cable clearance. Avoid 18 inch deep racks for this exact case.
- Unraid is not needed if the QNAP is the NAS. Use Ubuntu Server or Proxmox on PC 2 unless you replace the QNAP with a DIY NAS.

Rack Layout

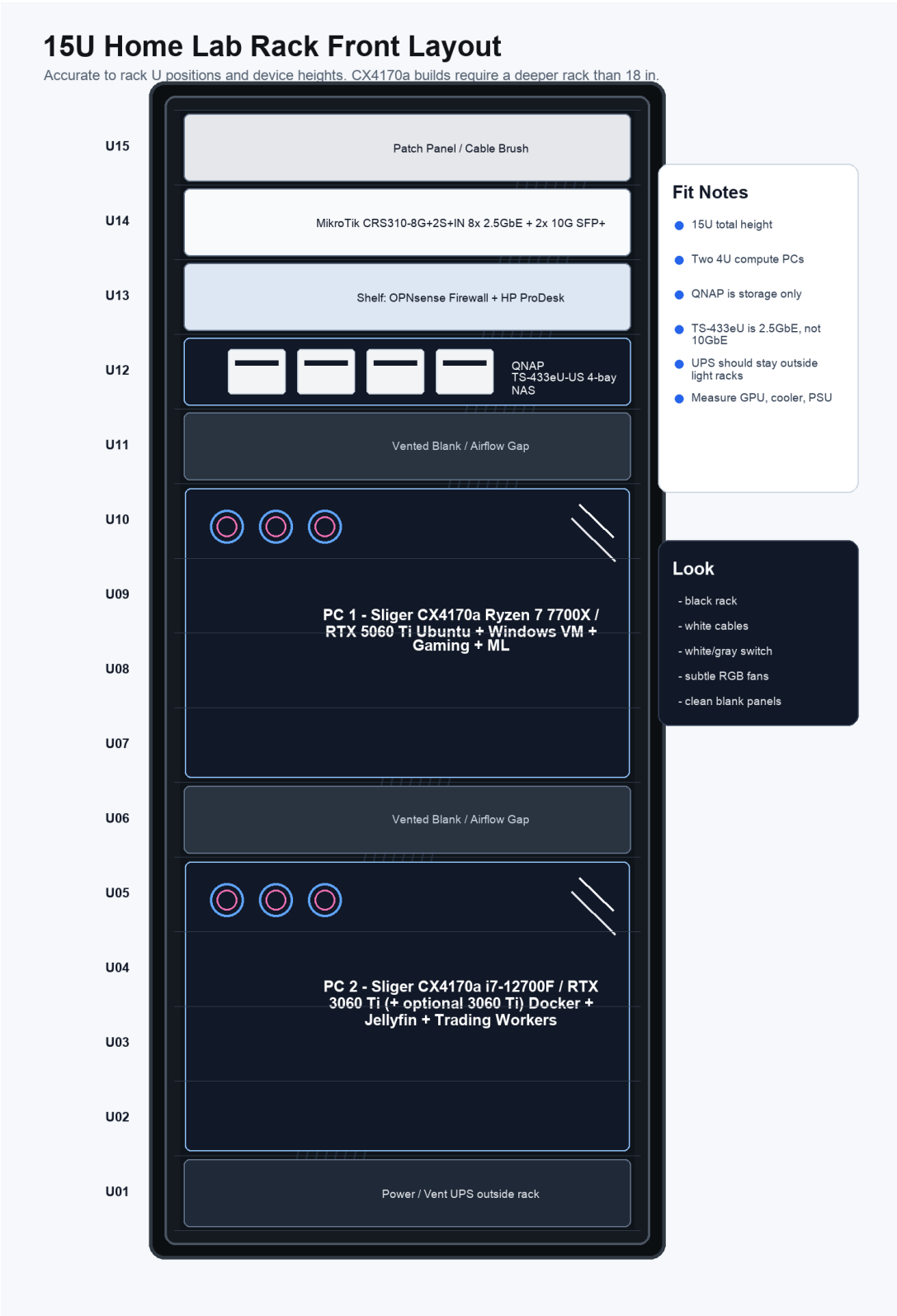


Figure 1. Front rack layout. U positions are accurate; product faceplates are schematic.

Rack U	Device	Role / Notes
U15	Patch panel / cable brush	White/black cable dress, short patch cables, labels.
U14	MikroTik CRS310-8G+2S+IN	8x 2.5GbE RJ45 and 2x 10G SFP+ for the local core.
U13	Shelf: OPNsense firewall + HP ProDesk	Firewall appliance plus the mini PC for always-on services.
U12	QNAP TS-433eU-US	1U 4-bay NAS for media, backups, shared datasets.
U11	Vented blank	Airflow gap above the hot compute box.
U10-U7	PC 1 in Sliger CX4170a	Ubuntu desktop, Windows VM, gaming, primary ML experiments.
U6	Vented blank	Airflow gap between the two compute PCs.
U5-U2	PC 2 in Sliger CX4170a	Docker/Jellyfin/trading services and second GPU compute node.
U1	Power / vent	Keep the UPS outside the rack unless the rack load rating is comfortably high.

Hardware Inventory And Upgrades

Device	Current Hardware	Role	Upgrade / Add
PC 1	Ryzen 7 7700X, ASUS PRIME B650M-A AX6 II micro-ATX, 32GB DDR5, 1TB SSD, RTX 5060 Ti	Ubuntu Desktop, Windows VM, gaming, ML/dev	Board fits CX4170a. Upgrade to 64GB DDR5 minimum; 128GB ideal if supported/stable. Add 2-4TB NVMe. Verify CPU cooler under 153mm for CX4170a.
PC 2	i7-12700F, MSI PRO B660M-A CEC WIFI DDR4 (MS-7D37) micro-ATX, 16GB DDR4, 1TB SSD, RTX 3060 Ti; optional second 3060 Ti	Ubuntu Server or Proxmox, Docker, Jellyfin, trading workers, secondary GPU jobs	Board fits CX4170a. Upgrade to 64GB DDR4 using matched DIMMs. Add 2-4TB NVMe. Dual GPUs are questionable on this micro-ATX board; check slot spacing, PSU, and airflow first.
HP ProDesk 400 G	i5-7500T, 28GB RAM, 480GB SSD	Always-on low-power services	Use as-is for AdGuard/Pi-hole, Tailscale, Uptime Kuma, and possibly Home Assistant.
QNAP TS-433eU-US	1U 4-bay NAS, ARM Cortex-A55, 4GB non-upgradeable RAM, dual 2.5GbE	Storage/media appliance	Use 2x16TB NAS drives to start if budget allows; 12TB drives are acceptable but fill a 4-bay NAS faster.
Firewall appliance	TBD	Router, firewall, VLANs, VPN	Add OPNsense box with 4x 2.5GbE ports. Protectli or Qotom-style hardware is fine.

Storage Plan

For a 4-bay NAS, 16TB drives are the cleaner long-term choice. 12TB IronWolf drives are fine if the deal is good, but four slots disappear quickly when media, backups, datasets, and model artifacts all live on the same NAS.

Stage	Drive Layout	Usable Capacity	Notes
Start	2 x 16TB RAID 1	~16TB	Safe starting point; one-drive failure tolerance.
Expand	3 x 16TB RAID 5	~32TB	More capacity, still one-drive failure tolerance.
Full	4 x 16TB RAID 5	~48TB	Best capacity use for this NAS; keep external/offsite backups.
Safer full	4 x 16TB RAID 6	~32TB	Two-drive failure tolerance, less usable space.
Existing drive	2TB Seagate	Do not add to main pool	Use as scratch, transfer, or offline backup. It would bottleneck a large RAID group.

Networking And Security

Layer	Recommended Part	Why
Firewall	OPNsense on a 4-port 2.5GbE appliance	Stable router/firewall, VLANs, WireGuard, DNS rules.
Switch	MikroTik CRS310-8G+2S+IN	Good compact core: 8x 2.5GbE and 2x 10G SFP+.
PC links	10GbE NICs + SFP+ DAC cables where needed	Use the 10G ports for PC 1 and PC 2. The QNAP remains 2.5GbE.
DNS/adblock	AdGuard Home or Pi-hole on HP ProDesk	Network-wide ad/tracker filtering and local DNS names.
Remote access	Tailscale or WireGuard	No open NAS/admin ports to the public internet.

Suggested VLANs: Main, Servers, Media, Trading/ML, IoT, Guest, and Management. Keep trading secrets and live trading services away from media/download containers.

Software Stack

Machine	Software	Notes
Firewall	OPNsense	WAN/LAN routing, VLANs, DHCP, firewall rules, WireGuard.
QNAP NAS	QTS, SMB shares, snapshots, QNAP backups	Storage only. Avoid exposing QNAP directly to the internet.
PC 1	Ubuntu Desktop, NVIDIA drivers, Docker, KVM/virt-manager, Windows 11 VM, Steam, Ollama	Primary workstation and gaming/ML box.

PC 2	Ubuntu Server or Proxmox, Docker Compose/Dockge, Jellyfin/Plex, trading containers	If Proxmox feels too much, start with Ubuntu Server.
HP ProDesk	Ubuntu Server, AdGuard Home/Pi-hole, Tailscale, Uptime Kuma	Low-power always-on utility node.

Missing Equipment Checklist

- 15U rack with enough depth and load capacity. CX4170a needs rear cable clearance; choose a deeper rack rather than the 18 inch VEVOR cabinet.
- 2x Sliger CX4170a cases, matching rails, and rack hardware for the compute PCs.
- RAM upgrades: PC 1 needs DDR5; PC 2 needs DDR4. Target 64GB each, with 128GB on PC 1 if the budget and board support make sense.
- 2-4TB NVMe SSD for PC 1 and 2-4TB NVMe SSD for PC 2.
- 2x16TB NAS HDDs to start; later expand to 4x16TB. If using 12TB drives, plan for lower ceiling.
- OPNsense firewall appliance with 4x 2.5GbE ports.
- MikroTik CRS310 switch, 10GbE NICs for the PCs if needed, SFP+ DAC cables, and white Cat6A patch cables.
- Vented blanks, brush panel, shelf, labels, Velcro ties, cable combs, and controlled RGB/white cable accents.
- UPS: CyberPower CP1500PFCRM2U or similar, preferably outside the rack if the cabinet load rating is low.

Fit Checks Before Buying

- Both known motherboards are micro-ATX, so both fit the CX4170a motherboard tray.
- PC 1 board: ASUS PRIME B650M-A AX6 II. Use DDR5 RAM only.
- PC 2 board: MSI PRO B660M-A CEC WIFI DDR4 (MS-7D37). Use DDR4 RAM only.
- Front-panel wiring: confirm each current case uses standard power/reset/LED connectors before transplanting.
- GPU exact model and length: CX4170a supports GPUs up to 375mm with front fans.
- GPU height including power cable: CX4170a supports up to 158mm including connectors.
- CPU cooler height: CX4170a air cooler limit is 153mm; use a compatible cooler or 360mm AIO.
- PSU length: keep at or under 190mm including cables.
- Rack usable depth: do not use an 18 inch deep cabinet with CX4170a; use a deeper rack or switch to CX4150a.
- Rack load rating: two 4U PCs plus NAS and shelves can approach light cabinet limits. Keep UPS outside unless the rack is rated for it.

Product Links

[Sliger CX4170a case](#)

[ASUS PRIME B650M-A AX6 II specs](#)

[MSI PRO B660M-A CEC WIFI DDR4 specs](#)

[QNAP TS-433eU-US](#)

[QNAP TS-433eU hardware specs](#)

[MikroTik CRS310-8G+2S+IN](#)

[Seagate IronWolf NAS drives](#)

[CyberPower CP1500PFCRM2U UPS](#)

[OPNsense install docs](#)

[Jellyfin Docker install](#)

Bottom Line

This build should be treated as a 15U aesthetic compute rack with a separate NAS appliance. The clean path is QNAP for storage, OPNsense for firewall, PC 1 for interactive Ubuntu/Windows/gaming/ML, PC 2 for Docker/media/trading workers, and HP ProDesk for low-power utility services.